

SmartAutomator - Adaptive API/UI Automation Framework

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Description

Intelligent automation framework that auto-selects between API calls and UI automation. Three script generation modes: prompt-based, screenshot-based, and recording-enhanced. No POM - AI dynamically generates resilient label-based XPath selectors. Includes humanizer layer and self-healing on selector failure.

Tech Stack

playwright, httpx, pydantic, click, rich, pillow, pyyaml, jinja2, pytest

Features

- API/UI hybrid strategy selector
- Prompt-to-script generation
- Screenshot-to-selector generation
- Recording enhancement with AI
- Self-healing selectors
- Humanizer (Bezier mouse, Gaussian delays, realistic typing)
- Cron scheduler
- Platform registry with YAML configs

Sample Input

Natural language automation request

```
smart-automator run --workflow linkedin_post.yaml --content 'Excited about my new project'
```

Sample Output

Automation execution results

```
Strategy: UI (no API available) | Generated 5 action steps | Executed in 12.3s | Status: SUCCESS
```

Project Files

```
.gitignore
LICENSE
project-idea.json
pyproject.toml
README.md
.github/workflows/test.yml
platforms/custom_template.yaml
```



```
tests/__init__.py
workflows/github_issue.yaml
workflows/linkedin_post.yaml
workflows/multi_platform.yaml
```

Test Results

PARTIAL

Run: 101 | Passed: 100 | Failed: 1

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